EKSWAR: Product Requirements Document (PRD)

Summary and language scope

EKSWAR dubs a short Hindi video into another Indian language in the creator's own cloned voice, with every line timed to the original and the background music kept.

Language scope. We plan to support all Indian languages. For this first version we added only three, Tamil, Bengali and Marathi, so we could build and test the whole pipeline well before widening it. The architecture does not depend on these three: adding a language means adding its language code, a translation prompt check and a native-speaker review, not a redesign.

	Languages
Source (v1)	Hindi
Target (v1, shipped)	Tamil, Bengali, Marathi
Target (planned)	All other Indian languages

Problem and opportunity

A Hindi creator reaches only part of India, which speaks hundreds of languages. Today's options each fail in one way:

- **Subtitles** get skipped on phones.
- **Human dubbing** is costly and replaces the creator's voice.
- **Automatic dubbing** drifts out of sync, because a translated line rarely takes as long to say as the original.

The opportunity is a dub that keeps the creator's voice and stays in sync, produced without re-recording.

Goals and non-goals

Goals

- Dub a Hindi video into Tamil, Bengali or Marathi in the creator's own voice.
- Keep every line in sync with where it was originally spoken.

- Keep the original background music.
- Check every dubbed line automatically and flag the ones that look wrong.
- Make re-runs cheap by caching every API result.

Non-goals for v1

- Lip-sync. EKSWAR matches timing only.
- Source languages other than Hindi.
- Videos with more than one speaker, or longer than 3 minutes.
- Replacing native-speaker review of translations.

Target users

Creators and organisations across India who want their videos to reach audiences in every regional language, without re-recording. That includes educators, small businesses, independent video makers, local brands, news and information channels, and training teams, all of whom cannot afford human dubbing for each language and want to keep their own voice.

The goal is for a creator in any part of India to publish once and be heard in the language of each region. Today the source is Hindi and the shipped targets are Tamil, Bengali and Marathi, with the remaining Indian languages planned.

Users work through a Streamlit web app or a command-line tool. They must confirm consent to clone the voice, and every output is labelled "AI-dubbed".

Scope: v1 and later

Area	v1 (built)	Later
Target languages	Tamil, Bengali, Marathi	All Indian languages
Source language	Hindi	More source languages
Speakers	One	Multiple
Length	Up to 3 minutes	Longer videos
Lip-sync	No, timing only	Separate stage
Translation review	Not yet done by native speakers	Native-speaker review per language
Interfaces	Streamlit web app, CLI	To be decided

Functional requirements

ID	Requirement	Status
F1	Accept a Hindi video and a voice sample of about 20 seconds	Done
F2	Let the user pick Tamil, Bengali or Marathi, or all three in one run	Done
F3	Separate voice from music and keep the music in the output	Done
F4	Transcribe Hindi with timestamps and split long lines at natural pauses	Done
F5	Translate each sentence with context, keep listed names, write numbers as words	Done
F6	Speak every line in the creator's cloned voice	Done
F7	Fit each line to its original time slot (timing agent), never cutting speech	Done
F8	Re-transcribe every dubbed line and flag scores below 80%	Done
F9	Output the dubbed MP4, an SRT per language and a `report.json`	Done
F10	Require consent before cloning a voice and label outputs "AI-dubbed"	Done
F11	Add more Indian languages	Planned

Non-functional requirements

- **Cost control:** every API result is cached on disk, so re-runs use no credits. The challenge budget is 5,000 credits.
- **Resilience:** a failed line never stops a run. Gnani 504 errors become "not checked" and are retried on the next run.
- **Safety:** inputs are checked before any credit is spent. Answers failing the script check are never cached.
- **Quality:** 173 automated tests, run offline.

- **Speed:** a cached re-run of the 27 s sample took 24 s. Demucs on CPU runs at about 0.8× real time.
- **Transparency:** the picture stream is copied unchanged, and outputs carry an "AI-dubbed" label.

Success metrics

Metric	Target	v1 result (2-minute demo, 3 languages)
Failed lines	0	0 of 63
Flagged lines	0	0 of 63
Recognition match on dubbed lines	80% or more	95–98%
Speaking-time coverage	High	89–92%
Automated tests passing	All	173 of 173

The 80% target comes from our quality check: a deliberately wrong line scored 36.7–47.5%, so 80% separates good lines from wrong ones.

Technical approach

The pipeline runs: audio prep, transcription, segmentation, translation, cloned-voice synthesis, timing agent, quality check, mix and export.

Component	Technology
Speech recognition	Gnani Prisma v2.5 (Batch for the transcript, REST for chunks and checks)
Voice cloning	Gnani `vachana-vc-v1`
Translation, rewriting	Google Gemini, with a backup model chain
Voice and music separation	Demucs 4.1.0
Audio and video	ffmpeg
Text similarity	rapidfuzz
Interface	Streamlit, plus a CLI

Adding a language means adding its code to the config, checking the translation prompt and script check for that language, confirming the cloned voice sounds acceptable, and getting a native speaker to review. The rest of the pipeline is language-neutral.

Risks and open questions

Risk	Impact	Mitigation
Translations not reviewed by native speakers	Wrong or unnatural lines	Native-speaker review before wider release
Cloned-voice quality may differ in new languages	Poor dub in a new language	Test each new language with the voice gate first; Timbre voices as fallback
Gemini overload or quota	Slow or failed runs	Backup model chain, caching, per-minute and per-day quota handling
Gnani REST 504 errors	Some checks show "not checked"	Retry on the next run
Voice cloning misuse	Someone's voice cloned without permission	Required consent box and "AI-dubbed" label

Open questions

- Which languages come after the first three, and in what order?
- Who reviews translations for each new language?
- Does the product need lip-sync, and at what cost?

Roadmap

1. **v1 (done):** Hindi to Tamil, Bengali and Marathi, with cloned voice, timing agent, quality check, web app and CLI. Submitted to the Gnani AI – Great Indian AI Internship Challenge.
2. **Next:** native-speaker review of the three languages.
3. **Then:** add the remaining Indian languages, one at a time, each through the same voice test and review.
4. **Later:** more source languages, multiple speakers, longer videos, and lip-sync.

The order after v1 is a plan, not a commitment; dates are not set.